BCSC0007: INTRODUCTION TO MICROPROCESSORS

Objective: *Objective of this subject is to introduce the basic concepts of microprocessor and assembly language programming. Identify and explain the operation of the components of typical microprocessor: the role of the ALU, registers, stack and the use of interrupts.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Introduction**: Microprocessors Evolution and Types, Basics of Pentium Microprocessor, Microprocessor Application, **8-Bit Microprocessor:**8085 Microprocessor and its Architecture, Addressing Modes, The 8085 Programming Model, Instruction Classification, Instruction Format, Overview of Instruction Set - Data Transfer Operation, Arithmetic Operation, Logic Operations and Branch Operations; Introduction to Assembly Language Program. **Programming Technique with Additional Instruction**: Looping, Counting, Indexing, Additional Data Transfer and 16-Bit Arithmetic Instruction, Counters and Time Delays, Stack and Subroutine.	20
II	**16 Bit Microprocessor:** Architecture of 8086 – Register Organization, Execution Unit, Bus Interface Unit, Signal Description, Physical Memory Organization, Mode of Operation, I/O Addressing Capabilities. **Peripheral Interfacing:** I/O Programming, Programmed I/O, Interrupt Driven I/O, DMA I/O, Memory-Mapped I/Os. **Peripheral Devices:** 8237 DMA Controller, 8255 Programmable Peripheral Interface, 8253/8254 Programmable Timer/Counter, 8259 Programmable Interrupt Controller.	18

Text Books:

- N Senthil Kumar, MSaravanan, and S Jeevananthan, "Microprocessors and Microcontrollers", Oxford University Press India, 2010.

Reference Books:

- Ramesh S. Gaonkar , "Microprocessor Architecture Programming and Applications with 8085", 4th Edition, Penram International Publishing, 2000.
- Ray A.K, Bhurchandi K.M, "Advanced Microprocessor and Peripherals", TMH, 2002.

Course Outcome

CO1- Demonstrate the Microprocessor internal architecture and its operations.
CO2- Develop programs based on 8085 microprocessor instruction set and addressing mode.
CO3- Develop program using looping, counting, indexing, counter and time delays.
CO4- Understand the concept of stack and subroutine for modular approach.
CO5- Compare accepted standards and guidelines to select microprocessor (8085 and 8086) to meet performance requirements.
CO6- Analyze the concept of interfacing the processor to external device with I/O programming and Interrupt Driven I/O.
CO7- Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259).

Printed Pages: **University Roll No. .....**

End Term Examination, Even Semester 2021-22
B.Tech (CSE/AIML/DA/IIOT/CCV/CSF), II-Year, IV-Semester
BCSC 0007: Introduction to Microprocessors

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. All parts of a question should be answer at one place.
2. First complete the one set of section then start a new section.
3. Answer should be brief and to-the-point.
4. Use suitable diagrams to explain the answer.
5. Assume the appropriate data for any missing data.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the need of branch group instructions? Explain the unconditional CALL instruction in detail.	4	2	U	C
2	Explain the function of following instructions: (a)LHLD 4000H (b) SPHL (c) OUT port address (d) DAD D	4	2	U	F
3	Write the four differences between 8085 and 8086 microprocessor.	4	5	R	C
4	Explain all the types of Indirect addressing modes of 8086 microprocessor. Also write suitable examples of each addressing mode.	4	2	U	F
5	Explain the concept of memory segmentation in 8086 microprocessor with suitable diagram. What are the advantages of memory segmentation in 8086 microprocessor? (OR) Draw the pin diagram of 8086 microprocessor and show all its pin signals.	4	1	R	P

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain all the control flags of 8086 Microprocessor.	3	1	U	F
2	Write an assembly language program for 10H bytes of data are stored in memory locations at 2050H to 205FH. Transfer this entire block of data to new memory location starting at 3050H.	3	3	A	P
3	Draw the internal architecture of 8259. Explain its key features.	3	7	U	C
4	Write an assembly language program to calculate sum of Odd numbers from a series. Length of the series stored in memory location 3000H and the series begins from memory location 3001H. Consider the sum to be 8 bit numbers. Store the sum at memory location 3500H.	3	3	A	P
5	Why microprocessor 8086 is faster than microprocessor 8085? How Pipelining is performed in 8086 microprocessor? Explain with the help of a diagram.	3	5	An	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Calculate the delay of given set of instructions. Assume the microprocessor is operating with the frequency of 2 megahertz. MVI B, 20H LOOP2: MVI C, FEH LOOP1: DCR C JNZ LOOP1 DCR B JNZ LOOP2	5	3	E	C
2	Draw the internal architecture of 8255. Mention the control word format of I/O and BSR Mode.	5	7	R	P
3	Explain the following signals of 8086 microprocessor pin diagram. (a) LOCK' (b) TEST' (c) BHE' (d) S0' S1' S2' (e) QS1, QS0	5	1	U	F

Printed Pages: 02

University Roll No.....

END -Term Examination, Even Semester 2018-19
B.TECH, (CS/DA/CCV), II YEAR, IV SEMESTER
BCSC 0007: Introduction to Microprocessor

Time: 3 Hour **Maximum Marks: 50**

Section – A

Attempt all questions **7 X 5 = 35 Marks**

Q1. Write a program in 8085 to find out the sum of numbers in a given series. The length of the series is in memory location 4200H and the series begins from memory location 4201H. Consider the sum to be 8 bit number. So, ignore carries. Store the sum at memory location 4300H.

Q2. Draw the pin diagram of 8086 microprocessor and explain the name of signal associated with maximum mode operation of 8086 microprocessor.

Q3. Sketch and explain the programming model of 8085.Also explain the need of demultiplexing of bus AD7-AD0 in 8085.

Q4. Explain mnemonic code with example

a. RAR b. DAA c. PUSH d. SHLD e. CPI Data

Q5. Highlight the use of addressing modes used in the following instruction.

a. ADI 23H **b.** MOV M, A **c.** LDA D500H
d. LXI D, 8C50H **e.** CMA

Q6. Draw the internal architecture of 8255. List the operating modes of 8255. Specify the bit of a control word for 8255, which differentiate between I/O mode and BSR mode?

Q7. Explain the purpose of peripheral interfacing in detail. What are the differences between interrupt driven I/O and DMA I/O?

OR

What is the necessity of the programmable interval timer? Discuss the organization of Timer with a functional block diagram.

Section – B

(a) **Attempt all questions** **3 X 2 =6 Marks**

Q1. How many 8259s IC can be connected in cascaded mode, show their Cascading structure?

Q2. Find the data direction and modes of operation of the 8255 ports, if the control word written into it is OAH.

Q3. Discuss the Command and Mode register configuration of 8237 DMA controller.

(b) **Attempt all questions** **3 X 3 =9 Marks**

Q1. Explain the data transfer types supported by 8237 and also explain the current and base register in 8237.

Q2. With the help of timing diagram explain mode0 and mode1 operation of 8253 timer/counter.

Q3. Use schematic diagram to discuss the internal architecture of PIC 8259.

END Term Examination, 2017-18

Course: B.Tech **Year: II** **Semester: IV**

Introduction to Microprocessor (CSE-4002)

Uni. Roll No: .....

Time: 2½ Hrs. **Total**

Marks: 40

Section A

Note: Attempt All Questions **(1x16=16)**

I. Which addressing modes are in given instruction?
   a. ADI 23H
   b. MOV M,A

II. What is the name of machine cycle if IO/M =1, S1=1 and S0=1?

III. How we increase the delay in any program and also explain the delay calculation of nested loop.

IV. Why interfacing is required for I/O devices?

V. How many address lines are necessary to address three megabyte (3072K) of memory?

VI. Explain RAR and DAA instruction with example?

VII. Explain the compare instruction with example in 8085 microprocessor.

VIII. What are the modes in which 8086 can operate?

IX. How is a 20 bit physical address is computed in 8086 microprocessor?

X. List the segment registers of 8086 microprocessor.

XI. What is the role of stack in calling a subroutine?

XII. Draw 8086 flag register format.

XIII. How many 8259 chip can be connected in cascaded mode?

XIV. What is the control word in BSR mode to set PC4?

XV. What are the priority options available in 8237A?

XVI. Highlight any two differences in 8253 and 8254.

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Draw and explain the bit pattern of mode register of 8237A.

II. Design and explain the each internal block diagram of 8255 PPI.

III. With proper required format explain the various operational command words of 8259 PIC.

IV. Write an ALP to blink port C bit 3 with 1 second delay of IC8255.

V. Explain all three I/O modes of operation in IC8255 and also gives some applications of IC8255.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. What are the different I/O modes of transfer explain each one? How DMA removed the disadvantages associated with the previous schemes.

II. Explain the 8254 control word format and set up the 8254 as a square wave generator with 1 ms period if the input frequency to 8254 is 1 MHz .Assume counter 0 is used to generate square wave and address of counter 0 = 3000H and control register =30006H.

III. Draw and explain the timing diagram of mode 2 operation of 8253 timer/counter, also explain the different modes of 8253.

IV. Draw the block diagram of Intel 8259 PIC. Also gives the initialization sequence of the 8259 ICW registers.

End Term Examination(2016-17)
Subject-Introduction to Microprocessor (CSE-4002)

Course: B.Tech.(CSE) **Year: II** **Semester: IV**
Duration: $2\frac{1}{2}$ Hrs **Maximum Marks-40**

Note:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Answer all Questions. $16 \times 1 = 16$

I. What are the advantages of memory segmentation in 8086 microprocessor?

II. List the four instructions which control the interrupt structure of the 8085 microprocessor.

III. An Intel 8085 A microprocessor is operated at a frequency of 2 MHz. If the instruction STA, 2100 H is executed, then what is the instruction cycle time ?

IV. How instruction cycle, machine cycle and T-states are related?

V. How many address lines in a 4096 x 8 EPROM CHIP?

VI. If the following program starts at 0100H, the contents of accumulator when PC reaches 0109H will be
LXI SP, 00FFH
LXI H, 0107H

MVI A, 20H
SUB M

VII. Write any four differences between 8085 and 8086?

VIII. What are the various DMA data transfer schemes?

IX. What is the significance of *READ BACK COMMAND* in 8254?

X. What is the purpose of control word written to control register in 8255?

XI. What is the difference between minimum and maximum modes of 8086?

XII. An 8085 assembly language program is given below. Assume that the carry flag is initially unset. The content of the accumulator after the execution of the program is

MVI A,07H
RLC
MOV B,A
RLC
RLC
ADD B
RRC

XIII. Draw and specify the complete bit configuration of 8085 flag Register?

XIV. Draw the control word of 8255 in I/O Mode.

XV. Total number of T States required to execute an instruction LXI H,1234H are..

XVI. How vectored interrupt is different from non vectored interrupt in 8085?

Answer any four Questions. $3 \times 4 = 12$

I. Explain control word of 8255 in BSR Mode. Consider the following addresses for PA=30H,PB=31, PC=32H, CWR=33H. Write a set of instructions to set PC0 and PC2 using BSR mode.

II. List the differences between 8253 and 8254.Write the name of different operating modes of 8253/54.

III. Differentiate between Memory mapped I/O and I/O mapped I/O.Draw and explain the control word of 8255 in I/O Mode.

IV. Write a set of instructions to Interface a key to 8255 and indicate its position on LED as shown in figure 1.
If switch is closed-LED should be ON
If switch is open-LED should be OFF
Assume Address of port A, PORT B and Control word is 30H , 31H and 32H respectively.

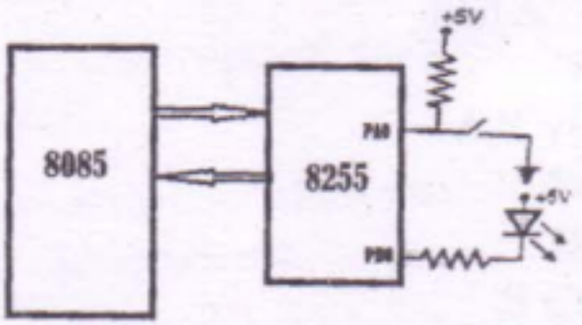


Figure 1:

V. Explain in brief different types of Data transfer scheme . Draw and explain mode set register and status register of 8237.

Answer any Three Questions. $4 \times 3 = 12$

I. Draw and explain the block diagram of Intel 8259 PIC.Also draw the flow chart for the initialization sequence of the 8259.

II. Explain the 8254 control word format and set up the 8254 as a square wave generator with 1ms period. The input frequency to the 8254 is 1 MHz. Assume counter 0 is used to generate square wave and address of counter 0=5000H and control register=50006H.

III. With the timing diagram and proper labelling explain in detail the mode 0 operation of 8253/54.Use counter 0 and count value is 05H.

IV. Draw and explain the internal block diagram of 8255 PPI. Write a set of instruction to initialize 8255 in the configuration given below

1. Port A : Output with handshake
2. Port B : Input with handshake
3. Port C_L : Output
4. Port C_U : Input
Assume address of the control word register of 8255 is 23H.

End Term Examination, 2015-16

Course: B. Tech. (CSE), Year: II, Semester: IV

Introduction to Microprocessor (CSE 4002)

Uni. Roll No:.....

Time: 150 Min. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What is the function of microprocessor in a system?

II. Differentiate Processor Cycle and Machine Cycle.

III. What is the name of machine cycle if IO/M =1, S1=0 and S0=1?

IV. What is the difference between microprocessor and microcontroller?

V. Which signal causes the microprocessor to immediately terminate its present activity?

VI. What is the addressing mode used in instruction MOV M, C?

VII. If the operating frequency is 5MHz then time required for 1 T-state is.....

VIII. Name the different types of flags used in 8085 microprocessor.

IX. What is interrupt service routine?

X. What do you mean by Non-Maskable interrupts?

XI. What is the position of the Stack Pointer after the POP instruction?

XII. Why interfacing is needed for 1/0 devices?

XIII. What is the control word in BSR mode to set PC4?

XIV. All the functions of the ports of 8255 are achieved by programming the bits of an internal register called as ....

XV. In cascaded mode, the number of vectored interrupts provided by 8259A is.....

XVI. Which microprocessor pins are used to request and acknowledge a DMA transfer?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Name the modes used by the DMA processor to transfer data and explain each transfer mechanism in short.

II. List the applications of programmable interval timer.

III. What is the purpose of control word written to control register in 8255?

IV. Explain in detail the operating modes of PPI? Also explain Interfacing.

V. Differentiate memory mapped I/O and I/O mapped I/O.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. Draw and explain the block diagram of 8254 programmable interval timer. Also explain the various modes of operation.

II. Discuss and explain interfacing of 8 bit input port with any another chip or processor in detail.

III. Draw and discuss the block diagram of the internal architecture of PIC 8259.

IV. What is the use of 8237 IC? Discuss various transfer modes of 8237.

Printed Pages: 03 **University Roll No.....**

End Term Examination, Even Semester, 2014-15
B.Tech. II Year, Semester-IV
CSE 222: Introduction to Microprocessor

Time: 02 ½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16 Marks)**

I. How many Total T-states required for LDA instruction? Justify your answer.

II. Which instruction is used for load 16 bit data (in Hexadecimal) to register pair? Write an example?

III. If value of CS register is 2000 and DS register is 4000 and value of IP is 5050. What will be physical address generated in 8086 microprocessor for fetching an instruction?

IV. Write the values of IO/M, S1 and S0 status pins for read cycle in 8085.

V. Define subroutine and stack. How they are related to each other? What is the use of CS pin in any chip?

VI. Which signal microprocessor is used to completely yield control of the bus to the DMA controller?

VII. What is the use of pre-fetch queue in 8086 microprocessor? Due to which causes the microprocessors immediately terminate its present activity?

VIII. What is the addressing mode used in instruction MOV M, C?

IX. How much time required for 5 T-state, if the given operating frequency is 5 KHz? Also show the calculation?

X. How many 8259 chip can be connected in cascaded mode?

XI. Write the control word of PPI chip for IO Mode where A port work as input port in Mode 2, B port work as output port in simple Input output mode and port C as input port.

XII. Write CWR contents for PIT chip for select the counter two, Read/Load MSB only in Square Wave Generator Mode for BCD counting.

XIII. Name the two input signals of 8254 programmable timer counter.

XIV. IRR stands for…

Note: Attempt any Four Questions **(3x4=12 Marks)**

I. Elucidate the operating modes of IC which is used for input output operation with 8085 microprocessor with example.

II. What is the necessity of the programmable interval timer? Discuss the organization of IC8254 PIT with a functional block diagram.

III. Explain the parallel I/O mode of 8255?

IV. Write the assembly language program to blink port C bit 3 with 1 second delay of IC8255.

V. Write short notes on following

a) Programmed I/O b) Memory Mapped I/O

Section C

Note: Attempt any Three Questions **(4x3=12Marks)**

I. Explain idle and active cycle DMA controller with diagram. What do you understand with modes of transfer? Explain DMA transfer with block diagram.

II. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

III. Explain 8259 (programmable interrupt controller) in detail. Explain different priority modes on 8259?

IV. Describe the organization of IC8237 with block diagram.

Printed Pages-3 **University Roll No.....**

B. Tech. II YEAR IV Semester, End Term Examination, 2013-14

CSE222: Introduction to Microprocessors

Time: 02 ½ Hours **Total Marks: 40**

Section- A

Note: Attempt All Questions. **1 x 16 = 16 marks**

I. Write the use of Direction Flag.

II. The no. of address lines required to address a memory of size 32 K is.....

III. A Bus cycle is equal to how many clock periods.

IV. Which microprocessor pins are used to request and acknowledge a DMA transfer?

V. What do you mean by Pseudo instructions?

VI. Which type of JMP instruction assembles if the distance is 0020 H bytes?

VII. The processor of knowing the status of device and transferring the data with matching speeds is called.....

VIII. What is Op-code?

IX. What are the names of the four segment registers?

X. Explain DAA instruction.

XI. In which condition over flow flag is set in 8086.

XII. Name the internal registers of 8086.

XIII. Write the two inputs signals of 8254 programmable timer counter.

XIV. IRR stands for....

XV. Which interrupts has highest Priority.

XVI. Name all the hardware interrupts?

Section- B

Note: Attempt Any Four Questions. **3 x 4 = 12 marks**

I. Write and explain the three data transfer mechanism in computer system.

II. Explain the parallel I/O mode of 8255?

III. Write and explain different mode of operation of 8253?

IV. Discuss the following mode of operation of 8259

i) Cascade mode ii) Buffered mode iii) Nested mode

V. Specify the number of times the following loops are executed. Also explain the flow of execution.

i)
```
        MVI  A, 17H
LOOP:   ORA  A
        RAL
        JNC  LOOP
```

ii)
```
        MVI  A, 17H
LOOP:   RAL
        ORA  A
        JNC  LOOP
```

iii)
```
        LXI  B, 1000H
LOOP:   DCX B
        NOP
        JNZ  LOOP
```

Section- C

Note: Attempt Any Three Questions. **4 x 3 = 12 marks**

I. Explain the data transfer types supported by 8237 and also write the bit pattern of command register of 8237A?

II. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

III. Explain the data transfer types supported by 8237 and also write the bit pattern of command register of 8237A?

IV. Discuss the organization of USART with a functional block diagram.

GLA University, Mathura **SET-A**

END Term Examination, 2012-13

Course: - B.Tech (CS) **II-Year, IV-Sem.**

Subject: - Introduction to Microprocessor (CSE 222) **Uni. Roll No:-**

Time:-2 hours 30 Minutes **Total Marks:-40**

Notes:-

1) *Answer all questions; Section A is **compulsory**, **Any four** from Section B and **Any four** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer): **(16*0.5 = 8)**

I. The act of acquiring an instruction is referred as the____ instruction.

- Fetching
- Fetch cycle
- Decode
- Execution

II. The subprogram finish the return instruction recovers the return address from the

- Queue
- Stack
- Program counter
- Pointer

III. The external system bus architecture is created using from _____ architecture:

- Pascal
- Dennis Ritchie
- Charles Babbage
- Von Neumann

IV. The processor 80386/80486 and the Pentium processor uses _____ bits address bus:

- 16
- 64
- 32
- 36

V. CS Stand for.

- Chip select
- Control select
- Cable select
- Computer select

VI. How many address ways of transferring data between the microprocessor and a physical I/O device?

- 1
- 2
- 3
- 4

VII. The CPU completes yields control of the bus to the DMA controller via:

- DMA acknowledge signal
- DMA integrated signal
- DMA implicitly signal
- None of these

VIII. Which are the parts of architecture of 8086?

- BIU
- 16 bit Accumulator
- EU
- All the above

IX. Which causes the microprocessor to immediately terminate its present activity?

- RESET signal
- INTERUPT signal
- RESET and INTERUPT signal
- None of these

X. INTR: it implies the_____ signal

- INTERUPT Request
- INTERUPT Reset
- INTERUPT Right
- INTERUPT wrong

XI. What is the control word in BSR mode to set PC4?

- 1xxx1000
- 0xxx1000
- 1xxx1001
- 0xxx1001

XII. Intel 8253/54 is a.

- USART
- DMA.
- TIMER
- Relative

XIII. If the operating frequency is 5MHz then time required for 1 T-state is:

- 0.1 µsec
- 0.2 µsec
- 0.3 µsec
- 0.4 µsec.

XIV. What is the addressing mode used in instruction MOV M,C?

- Direct
- Indirect
- Immediate
- Indexed

XV. CMP R/M instruction of 8085 microprocessor set zero flag

- A > R/M
- A=R/M
- A < R/M
- None of these

XVI. 8254 can handle input upto

- 5MHz
- 10MHz
- 15MHz
- 20MHz

SECTION-B (Attempt any four) (4*3=12)

I. Write a program to add the contents of the memory location 2000H: 0500H with contents of 3000H: 0600H memory location and store the result in 5000H: 0700H?

II. What is need of addressing modes in microprocessor? Give overview of microprocessor family.

III. Explain the following instructions with suitable example. Also indicate the flag condition.
i) ADI ii) DAA iii) RAR

IV. List the internal registers of 8086 microprocessor with their abbreviations.

V. Discuss relationship between Instruction cycle, Machine cycle and T-state with diagram.

SECTION-C (Attempt any four) (4*5=20)

I. Draw and discuss the block diagram of the internal architecture of PIC 8259.

II. What is the necessity of the programmable interval timer? Discuss the organization of Timer with a functional block diagram.

III. What is the use of 8237 IC? Discuss mode register and various modes of 8237.

IV. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

V. Write short notes on it,
i) Programmed I/O ii) Memory Mapped I/O